STAT 330 Practice Exam 1

Modules 1, 2, and 3

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle all that apply) A multiple regression model has $n = 84$ observations and $p = 4$ predictors, plus an intercept. Which statements are correct?

(A) The design matrix X is 84×5.

(B) The coefficient vector $\boldsymbol{\beta}$ is 4×1.

(C) The fitted-value vector $X\widehat{\boldsymbol{\beta}}$ is 84×1.

(D) The hat matrix $H = X(X^\top X)^{-1}X^\top$ is 84×84.

(E) The residual vector is 5×1.

Question 2: (TRUE/FALSE) A theoretical regression model contains unknown population parameters and an error term, while a fitted equation contains estimated coefficients and no separate error term.

(A) TRUE

(B) FALSE

Question 3: (Circle one answer) Ordinary least squares chooses $\widehat{\boldsymbol{\beta}}$ to minimize

(A) $\sum_i e_i$.

(B) $\sum_i |e_i|$.

(C) $\sum_i e_i^2$.

(D) $\sum_i (y_i - \bar{y})^2$.

Question 4: (Circle all that apply) For the same predictor values and confidence level, which statements about confidence and prediction intervals are correct?

(A) A prediction interval for one new outcome is usually wider than a confidence interval for the mean outcome.

(B) A confidence interval for the mean is designed to contain one specific future outcome.

(C) Both intervals are centered at the same fitted value.

(D) A prediction interval includes uncertainty about the mean and individual-to-individual variation.

(E) The two intervals always have the same standard error.

Question 5: (Circle all that apply) Which statements about the LINE assumptions and their consequences are correct?

(A) Independence must be judged partly from how the data were collected.

(B) A small Shapiro-Wilk p-value automatically means the model must be discarded.

(C) Heteroscedasticity can make usual standard errors, intervals, and tests unreliable even when coefficient estimates remain unbiased.

(D) Curvature in an added-variable plot can indicate a linearity problem after accounting for other predictors.

(E) Any observation with Cook's distance above $4/n$ must be deleted.

Question 6: (Circle all that apply) Which statements about model fit are correct?

(A) Ordinary R^2 cannot decrease when another predictor is added to an OLS model fit to the same observations and response.

(B) Adjusted R^2 can decrease after a weak predictor is added.

(C) A significant overall F-test provides evidence that at least one slope differs from zero.

(D) RMSE values can always be compared directly when two models use different transformations of the response.

(E) The model with the largest training-set R^2 is guaranteed to have the smallest future prediction error.

Question 7: (Circle one answer) Suppose

$$\log(y) = \beta_0 - 0.035x_1 + \beta_2 x_2 + \epsilon.$$

Which interpretation of the x_1 coefficient is most appropriate?

(A) Holding x_2 constant, a one-unit increase in x_1 is associated with a 0.035-unit decrease in average y.

(B) Holding x_2 constant, a one-unit increase in x_1 is associated with about a 3.4% decrease in the conditional median of y.

(C) A 1% increase in x_1 is associated with a 3.5% decrease in y.

(D) Increasing x_1 by one unit causes every value of y to fall by exactly 3.5%.

Question 8:

(a) Describe one complete repetition of repeated train-test validation, beginning with the split and ending with a recorded test-set RMSE.

(b) What should be done with the recorded RMSE values after all repetitions are complete?

(c) Explain why candidate models should use the same train-test splits.

Question 9: (Circle all that apply) Which of the following statements about leverage and Cook's distance are true?

(A) An observation can have high leverage but a small Cook's distance if its response lies close to the fitted regression surface.

(B) Cook's distance summarizes how much the fitted values change when an observation is removed and the model is refit.

(C) Any observation with Cook's distance greater than $4/n$ should automatically be removed from the dataset.

(D) An observation with both high leverage and a large standardized residual will tend to have a large Cook's distance.

(E) The leverage of an observation depends on its observed response value y_i.

Question 10:

(a) Let $\boldsymbol{b}$ be a candidate coefficient vector, meaning any proposed value for $\boldsymbol{\beta}$. It could be $\widehat{\boldsymbol{\beta}}$ or some other guess. For this candidate $\boldsymbol{b}$, write the residual vector and the residual sum of squares in matrix notation.

(b) Explain conceptually how ordinary least squares uses this residual sum of squares to estimate the regression coefficients.

(c) Explain why $\widehat{\boldsymbol{\beta}}$ can change from sample to sample even though the population coefficient vector $\boldsymbol{\beta}$ is fixed.

PART II

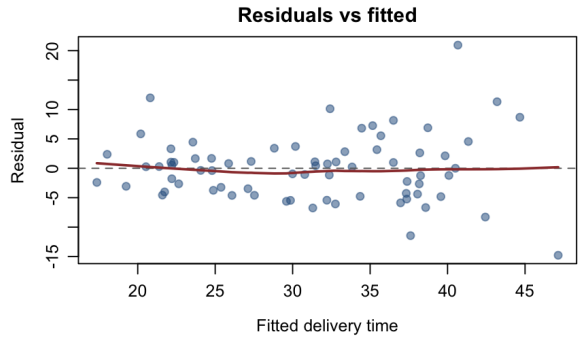
An urban logistics company studies the duration of 72 package-delivery routes. Each observation comes from a different driver-day combination. The variables are route duration in minutes (`duration`), route distance in kilometers (`distance`), and number of delivery stops (`stops`). The data are observational.

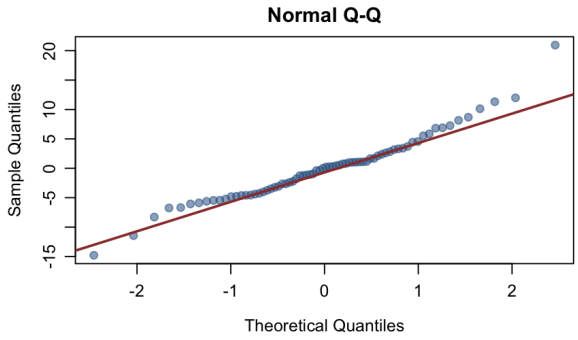
Model 1: Untransformed response

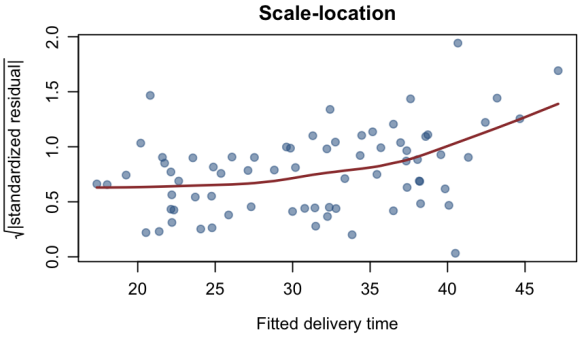
$$\texttt{duration}_i = \beta_0 + \beta_1 \texttt{distance}_i + \beta_2 \texttt{stops}_i + \epsilon_i, \qquad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2).$$

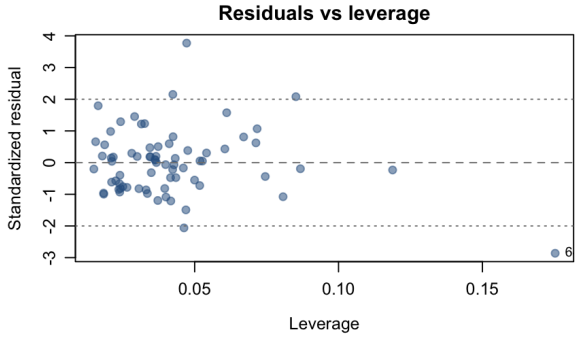
Coefficient	Estimate	Std. Error	t value	p-value
Intercept	13.670	1.913	7.146	< 0.0001
`distance`	0.888	0.127	6.997	< 0.0001
`stops`	0.234	0.178	1.319	0.1915

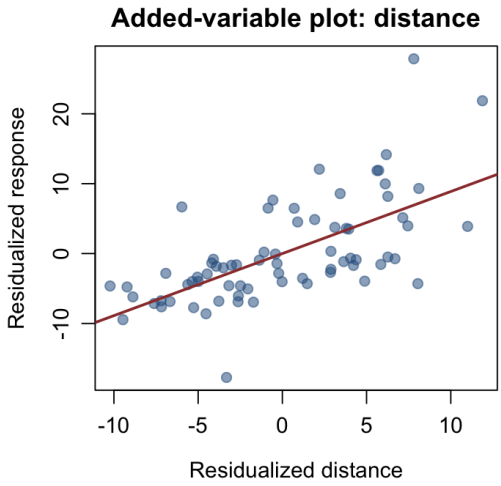
Other summary	Value
Breusch-Pagan p-value	0.007
Shapiro-Wilk p-value	0.0240
Repeated-CV RMSE, minutes	5.94
Cook's distance, route 64	0.580

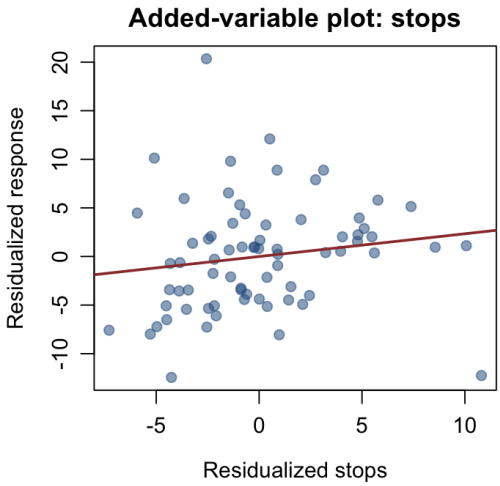












Question 11: State the estimated value of β_1 and interpret it in the context of the routes.

Question 12: Use $t^* = 1.995$ to construct a 95% confidence interval for the coefficient on `stops`. Show the calculation, then state what the interval implies about a two-sided test of $H_0 : \beta_2 = 0$ at $\alpha = 0.05$.

Question 13: Assess each of the following for Model 1 using both the plots and information about data collection. Treat diagnostic evidence as a matter of degree.

(a) Linearity

(b) Independence

(c) Normality

(d) Equal variance

(e) Unduly influential observations

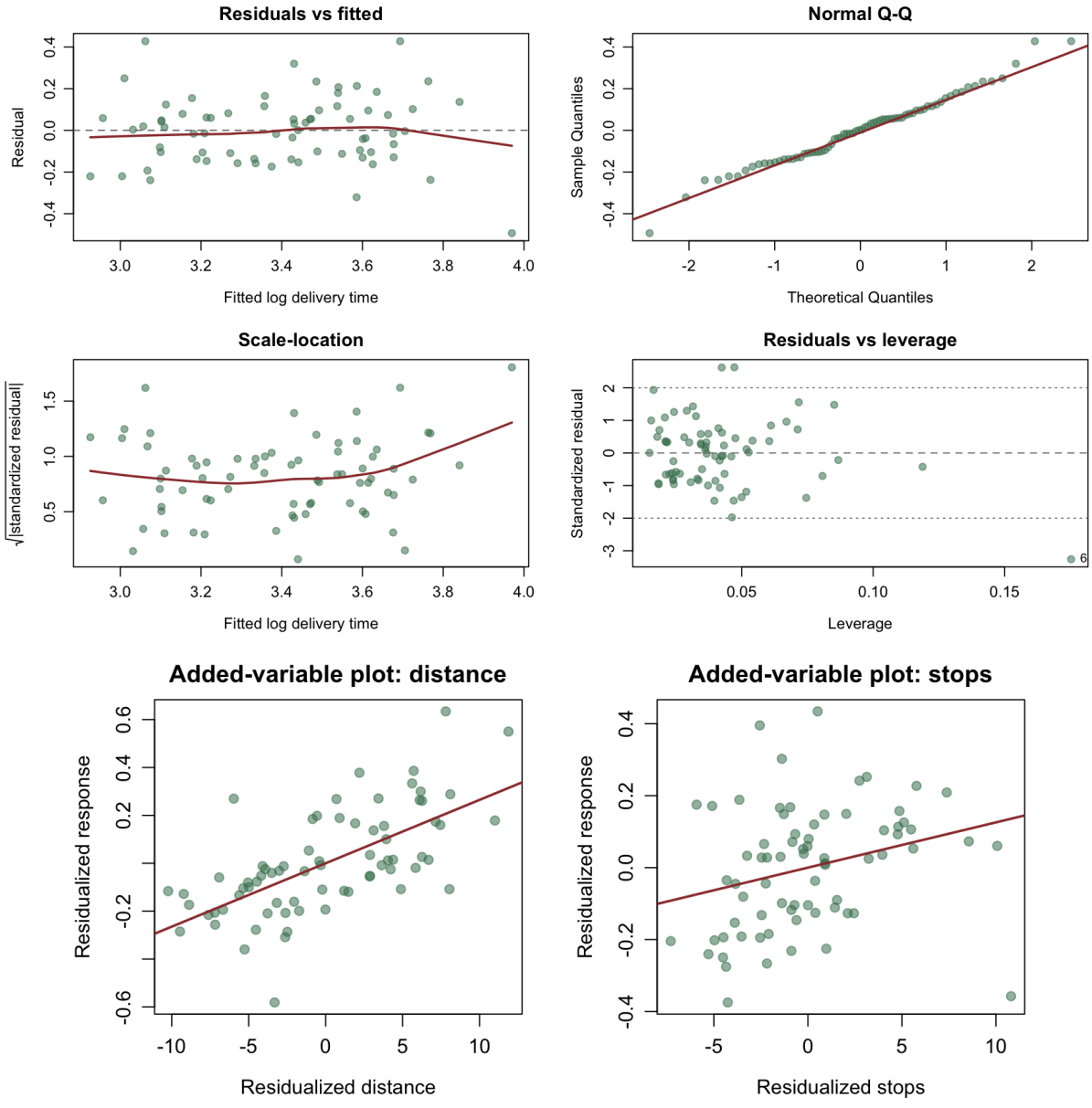
Model 2: Log-transformed response

$$\log(\texttt{duration}_i) = \beta_0 + \beta_1 \texttt{distance}_i + \beta_2 \texttt{stops}_i + \epsilon_i, \qquad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2).$$

Coefficient	Estimate	Std. Error	t value	p-value
Intercept	2.799	0.0560	49.966	< 0.0001
`distance`	0.0265	0.0037	7.121	< 0.0001
`stops`	0.0126	0.0052	2.418	0.0183

Other summary	Value
Breusch-Pagan p-value	0.288
Shapiro-Wilk p-value	0.4681
Repeated-CV RMSE, minutes	6.05
Cook's distance, route 64	0.753

The cross-validation predictions for Model 2 were exponentiated before RMSE was calculated, so both reported RMSE values are on the original minute scale.



Question 14: Route 64 exceeds the $4/n$ Cook's-distance guideline in both models. Explain what this tells you and describe an appropriate next step. Be explicit about what the guideline does not justify.

Question 15: Compare Models 1 and 2. State the evidence favoring each model, then give a defensible recommendation. Do not decide from one p-value alone.

Question 16: Interpret the `distance` coefficient in Model 2. Give both the exact percentage-change expression and an appropriate approximation.

Question 17: For a route of 18 kilometers with 14 stops, Model 2 gives a fitted log duration of 3.452. Calculate the predicted conditional median duration on the original scale.

Question 18:

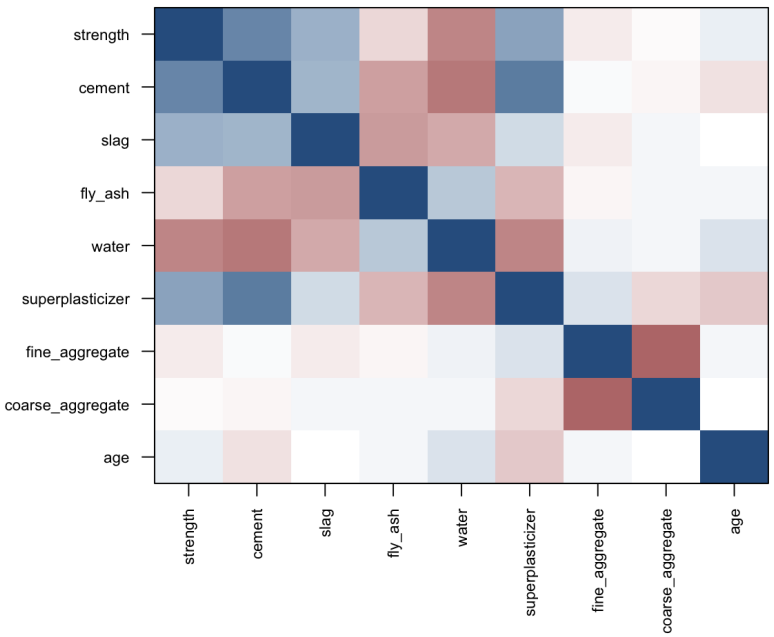
(a) What degrees of freedom should be used for a t critical value in Model 2?

(b) The company wants an interval for the mean log duration among all routes with 18 kilometers and 14 stops. Should it use a confidence interval for the mean or a prediction interval for one new route? Briefly explain.

Question 19: For routes with 18 kilometers and 14 stops, the standard error for the fitted mean on the log scale is 0.021 and $t^* = 1.995$. Construct a 95% confidence interval for the mean log duration, exponentiate its endpoints, and interpret the resulting interval on the original scale.

PART III: APPLICATION

A materials laboratory studies 96 concrete batches. The response is compressive strength in megapascals (`strength`), measured at the recorded curing age. The mixture predictors measured in kilograms per cubic meter are `cement`, `slag`, `fly_ash`, and `water`. The two aggregate predictors, `fine_aggregate` and `coarse_aggregate`, use the same units. The remaining predictors are `superplasticizer`, in liters per cubic meter, and curing `age`, in days. The laboratory is comparing explanatory stability, parsimony, and out-of-sample prediction.



The largest VIF in the full eight-predictor model belongs to `cement`. In the auxiliary regression of this predictor on the other seven predictors, $R_j^2 = 0.598$.

Question 20:

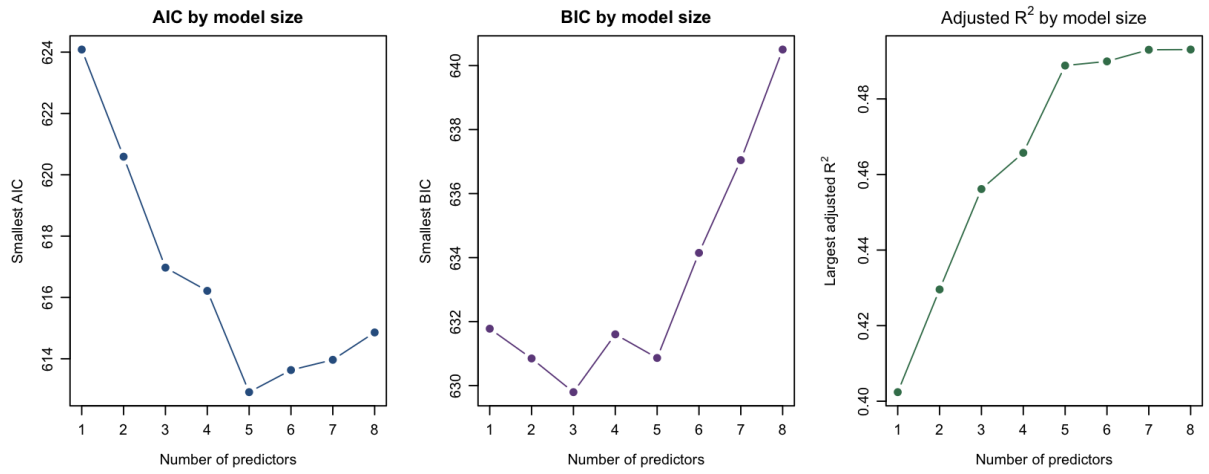
(a) Calculate the VIF for `cement` using $\text{VIF}_j = 1/(1 - R_j^2)$.

(b) Calculate the factor by which its coefficient standard error is inflated, relative to an otherwise comparable orthogonal design.

(c) Briefly interpret both quantities. Does this calculation prove that the predictor should be removed?

Best-subsets search produced the following winners. The row label names the metric used to choose the model.

Criterion	Selected predictors	Criterion value
AIC	`cement + slag + fly_ash + water + age`	612.9
BIC	`cement + water + age`	629.8
Adjusted R^2	all eight predictors	0.493



Question 21:

(a) Distinguish a model-search algorithm from a model-selection metric. Classify best subsets and BIC using those terms.

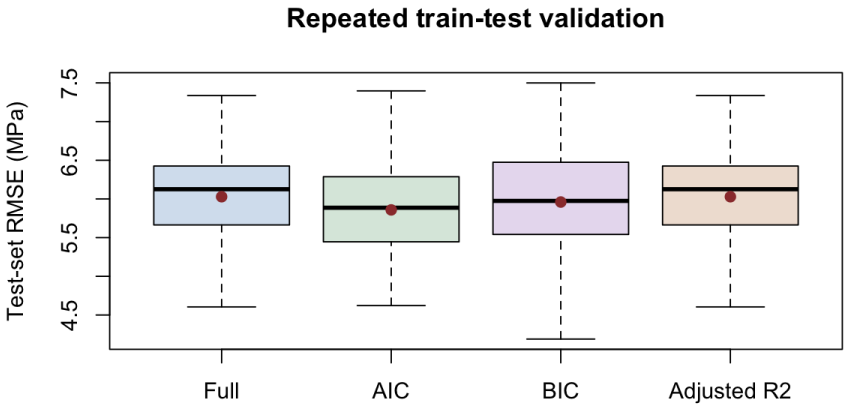
(b) Why can AIC, BIC, and adjusted R^2 select different models from the same data?

(c) Which result above is most consistent with a stronger penalty for model complexity?

Question 22: Suppose forward selection chooses `superplasticizer` as its first one-predictor model, but the best two-predictor model under a chosen metric is `cement + water`. Explain why forward selection cannot reach that two-predictor model on its next step, even if best subsets can find it.

The same 120 train-test splits were used for four candidate models. The points in the plot mark mean test-set RMSE.

Model	Full	AIC winner	BIC winner	Adjusted-R^2 winner
Mean RMSE (MPa)	6.03	5.86	5.96	6.03



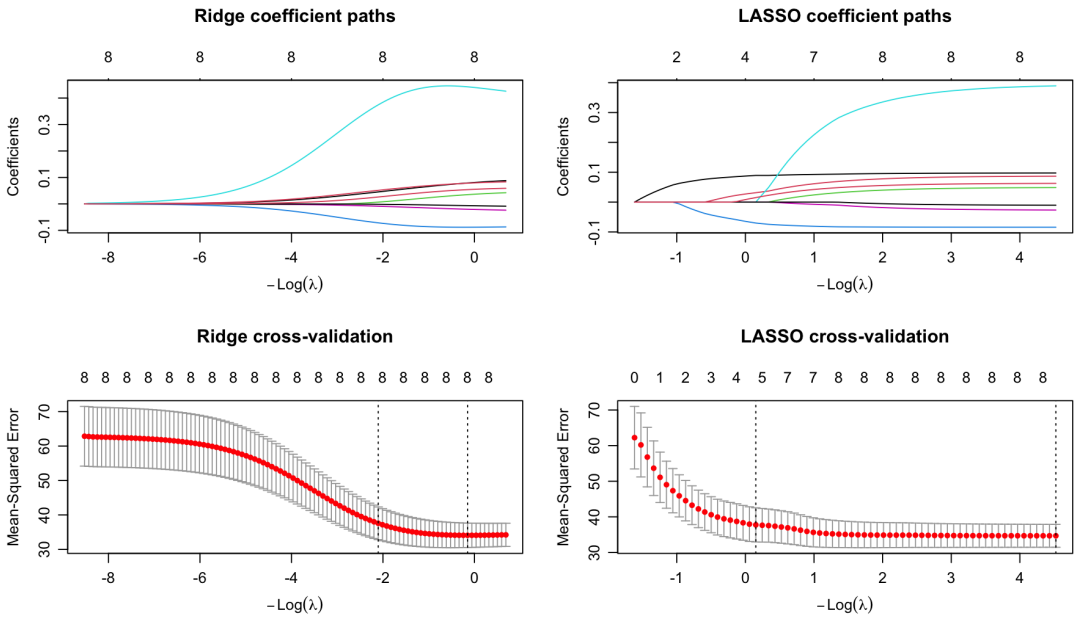
Question 23:

(a) Which model has the smallest mean CV RMSE, and what would you tentatively choose if prediction were the only goal?

(b) The differences are small relative to the split-to-split variation. Describe how paired RMSE differences should be used before declaring one model superior.

(c) Explain why the full model can fit the training data well without winning this comparison.

Ridge and LASSO were fit after standardizing the predictors. Their coefficient paths and 10-fold cross-validation curves are shown below.



For LASSO, $\lambda_{\min} = 0.011$ retains 8 predictors, while $\lambda_{1se} = 0.859$ retains 4 predictors: `cement`, `slag`, `water`, `age`.

Question 24:

(a) **(Circle all that apply)** Which statements are correct?

(A) Ridge usually shrinks correlated coefficients without setting them exactly to zero.

(B) LASSO can produce a sparse model by setting some coefficients exactly to zero.

(C) Both methods intentionally introduce bias that can reduce variance and improve prediction.

(D) A larger λ means a weaker penalty and less shrinkage.

(E) Classical OLS standard errors automatically remain valid for coefficients selected by LASSO.

(b) If the laboratory prioritizes the smallest estimated CV error, which LASSO value of λ should it use? If it instead wants a simpler model whose error is within one standard error of the minimum, which value should it use?

(c) Recommend ridge or LASSO when the laboratory believes many correlated ingredients each contribute a little. Then recommend a method when it needs a short list of ingredients for communication.

Scratch Paper

Scratch Paper

Scratch Paper

Scratch Paper